Anatomy & Physiology — Body Systems Sheets

Structure–function tables, physiology essentials, clinical links and exam traps, system by system

MindShelf

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How to Use These Sheets

Each of the 11 body systems follows the same four-part layout:

1. **Key structures → function:** a table that pairs each structure with what it does.
2. **Physiology essentials:** the mechanisms behind the table.
3. **Clinical link:** one condition that shows the physiology at work, with the source it comes from.
4. **Traps:** mix-ups that cost points on exams.

These sheets summarize core facts for review. They are not a full course and do not replace your textbook or your program's materials.

Levels of organization

Cells (the smallest units that can live and reproduce) → **tissues** (groups of similar cells and the material between them) → **organs** (several tissues working together, e.g., the stomach) → **organ systems** (organs that share a complex job).

Directions and planes

Term	Meaning
Superior (cranial) / inferior (caudal)	Toward the head / away from the head
Anterior (ventral) / posterior (dorsal)	Front / back of the body
Medial / lateral	Toward the midline / away from the midline
Proximal / distal	Closer to / farther from the trunk or a limb's point of origin
Sagittal plane	Vertical, front to back: splits right from left (median plane = exactly on the midline)
Coronal (frontal) plane	Vertical, side to side: splits anterior from posterior
Transverse (axial) plane	Horizontal: splits upper from lower parts

Integumentary System

Key structure	Function
Epidermis	Outer layer of keratinocytes; it has no blood vessels of its own and is nourished from the dermis
Stratum basale	Deepest epidermal layer: dividing basal cells, melanocytes (make melanin) and Merkel cells (touch)
Stratum spinosum	Keratinocytes plus Langerhans cells, which pick up antigens entering damaged skin and alert the immune system
Stratum granulosum, lucidum	Keratinocytes flatten and lose water as they are pushed toward the surface; the lucidum exists only in thick skin (palms, soles, digits)

Key structure	Function
Stratum corneum	Surface layer: 10 to 30 rows of dead keratinocytes forming the tough outer barrier
Dermis, papillary layer	Thin collagen fibers and small vessels that feed the epidermis and help regulate temperature
Dermis, reticular layer	Thicker, denser collagen: strength, structure and elasticity
Dermal structures	Hair follicles, sebaceous (oil) glands, nerve endings, blood vessels; arrector pili muscles raise goosebumps. Sweat glands lie deep, down to the subcutaneous layer, and open on the surface through pores
Hypodermis (subcutis)	Network of fat and collagen crossed by blood vessels, lymph vessels and nerves

Physiology essentials

- **Functions:** barrier against pathogens, injury, heat and light; temperature control; sensation; storage of water and fat; vitamin D; a front line for the immune system.
- **Renewal:** new cells form in the stratum basale and are pushed upward, flattening and dying until they reach the stratum corneum.
- **Pigment:** melanocytes in the basal layer make melanin, which helps protect the skin from damaging UV rays.
- **Heat:** sweat reaches the surface through pores and cools the body as it evaporates; in the cold, dermal blood vessels constrict to keep heat.
- **Vitamin D:** with UV light the epidermis turns a cholesterol derivative into vitamin D_3; the liver and kidneys activate it, and it is needed to absorb calcium and phosphorus for bone.
- **Sensation:** Merkel cells and dermal receptors sense touch, temperature and pain; cells around hair follicles sense pressure.

CLINICAL NOTE Pressure injuries
Pressure injuries (bedsores) form when a person stays in one position too long, most often where bone lies close to the skin: heels, hips, ankles, elbows, back. People who are bedridden, use a wheelchair or cannot shift on their own are most at risk. Prevention: keep skin clean and dry, change position about every 2 h, use pressure-relieving cushions and surfaces. Source: MedlinePlus, *Pressure Sores* (consulted Oct. 29, 2026).

TRAP
The epidermis has **no blood vessels**. A scrape that bleeds has reached the dermis.

TRAP
Melanocytes sit in the **stratum basale** of the epidermis, not in the dermis.

TRAP
Deep to surface: basale → spinosum → granulosum → lucidum → corneum. Thin skin has only 4 layers (no lucidum), thick skin 5. Questions often list them surface first.

Skeletal System

Key structure	Function
Compact bone	Dense bone built of osteons (haversian systems): a central canal with blood vessels ringed by lamellae of matrix
Lacunae and canaliculi	Small spaces that hold osteocytes, linked by tiny channels that carry nutrients
Spongy (cancellous) bone	Trabeculae (plates and bars) that follow lines of stress; the spaces between them hold red marrow
Red marrow	Hematopoiesis: forms blood cells
Osteoblasts	Form new bone
Osteocytes	Mature bone cells within the matrix
Osteoclasts	Resorb (break down) bone; their balance with osteoblasts maintains bone tissue
Axial skeleton	80 bones: skull (with the ear ossicles), hyoid, vertebral column, thoracic cage (sternum and ribs)
Appendicular skeleton	126 bones: pectoral girdle (clavicles, scapulae), upper limbs, pelvic girdle (hip bones), lower limbs
Synarthroses	Immovable joints, bones held by thin fibrous tissue (e.g., skull sutures)
Amphiarthroses	Slightly movable joints joined by cartilage (e.g., pubic symphysis, intervertebral disks, costal cartilages)
Diarthroses (synovial)	Freely movable, the most common adult joints: articular cartilage, joint cavity, fibrous capsule with ligaments, synovial membrane making lubricating fluid

Physiology essentials

- **Functions:** support against gravity; protection of the brain, spinal cord, heart and lungs; levers for movement with the muscles; calcium storage and release; blood cell formation in red marrow.
- **Count:** 80 axial + 126 appendicular = 206 bones in the adult skeleton.
- **Living tissue:** bone needs a blood supply, uses oxygen and nutrients, and remodels under mechanical stress; trabeculae realign when the direction of stress changes.
- **Calcium:** calcium is the most abundant mineral of the matrix. PTH raises blood Ca^{2+} and calcitonin lowers it (see Endocrine System).
- **Remodeling:** osteoblasts lay down bone and osteoclasts resorb it; the balance between them keeps bone mass stable.

CLINICAL NOTE Osteoporosis

Bones weaken and break more easily. It is usually silent until a fracture, most often of the hip, a vertebra or the wrist. Risk rises with female sex (especially after menopause), age over 50, a slim build, family history, long-term use of some medicines such as corticosteroids, smoking, heavy drinking and inactivity. Prevention relies on calcium and vitamin D in the diet and regular physical activity. Source: MedlinePlus, *Osteoporosis* (consulted Oct. 29, 2026).

TRAP

Osteoblasts build, **osteoclasts** break down, **osteocytes** are the mature cells that stay in the matrix.

TRAP

The pectoral and pelvic **girdles** belong to the appendicular skeleton, not the axial one.

TRAP

Most adult joints are freely movable **synovial** joints; skull sutures are immovable (synarthroses).

Muscular System

Key structure	Function
Skeletal muscle	Striated, voluntary, many nuclei per fiber; attached to bones
Cardiac muscle	Striated, involuntary (autonomic control), one central nucleus per cell; wall of the heart
Smooth muscle	Not striated, involuntary, one central nucleus; walls of hollow organs and vessels (gut, bladder, uterus)
Connective sheaths	Epimysium wraps the whole muscle, perimysium each fascicle (bundle of fibers), endomysium each fiber
Tendon and aponeurosis	Extensions of the sheaths that attach a muscle to the periosteum of a bone or to another muscle
Neuromuscular junction	The motor neuron releases acetylcholine (ACh); the fiber depolarizes and the impulse runs into the T-tubules
Sarcoplasmic reticulum	Stores Ca^{2+}, releases it when the fiber is stimulated, then pumps it back (using ATP) so the fiber relaxes
Troponin and tropomyosin	Ca^{2+} binds troponin, which shifts tropomyosin off the binding sites on actin
Sarcomere (actin and myosin)	Myosin heads bind actin and pull the thin filaments toward the center: the sarcomere shortens, the filaments do not

Physiology essentials

- **Functions:** movement of bones, organs and vessels; posture; stability of the joints the tendons cross (knee, shoulder); heat, with nearly 85% of body heat coming from muscle contraction.
- **Excitation:** a fiber contracts only after a nerve impulse: ACh at the neuromuscular junction → action potential along the membrane and T-tubules → Ca^{2+} release.
- **Cross-bridge cycle:** Ca^{2+} on troponin uncovers actin → myosin heads attach and pull (power stroke) → ATP lets them let go and re-cock → the cycle repeats while Ca^{2+} and ATP last.
- **Relaxation:** ACh release stops, Ca^{2+} is pumped back into the sarcoplasmic reticulum, tropomyosin covers actin again.
- **Energy:** creatine phosphate regenerates ATP for the first seconds of intense work; glycolysis covers short intense bursts; aerobic respiration supplies most of the ATP at rest or in moderate activity.

CLINICAL NOTE Myasthenia gravis
An autoimmune disease: antibodies disrupt signal transmission at the neuromuscular junction, so muscles do not get the nerve signal. Weakness of voluntary muscles worsens with activity and improves with rest; drooping eyelids and double vision are often first. Watch chewing and swallowing, and breathing, since chest-wall weakness can impair it. Source: MedlinePlus, *Myasthenia gravis* (consulted Oct. 29, 2026).

TRAP
ATP is needed to **release** myosin from actin, not only for the power stroke. No ATP means locked cross-bridges (rigor mortis), not relaxation.

TRAP
During contraction the **sarcomere** shortens; actin and myosin filaments keep their length and slide.

TRAP
Striated does not mean voluntary: cardiac muscle is striated **and** involuntary.

Nervous System and Special Senses

Key structure	Function
CNS	Brain and spinal cord, in the dorsal body cavity, protected by the skull and vertebrae
PNS	12 pairs of cranial nerves and 31 pairs of spinal nerves; every spinal nerve is mixed (sensory and motor)
Neuron	Dendrites carry impulses toward the cell body; the single axon carries them away to the synaptic bulbs
Myelin	Made by Schwann cells (PNS) and oligodendrocytes (CNS); gaps are nodes of Ranvier; myelinated fibers form white matter
Neuroglia	Support, nourish and protect neurons; far more numerous, and unlike neurons they can divide
Meninges	Dura mater (tough, outer), arachnoid (cerebrospinal fluid in the subarachnoid space below it), pia mater (on the surface)
Cerebrum	Two hemispheres joined by the corpus callosum. Frontal: motor cortex, planning, Broca's area (speech production); parietal: touch and body sense; temporal: hearing, long-term memory; occipital: vision
Diencephalon	Thalamus relays sensory input except smell to the cortex; hypothalamus runs homeostasis, the autonomic system and the pituitary
Brainstem	Midbrain, pons, medulla oblongata; the medulla continues as the spinal cord and carries all ascending and descending tracts
Cerebellum	Coordination: compares motor commands with sensory feedback and sends corrections
Spinal cord	From the foramen magnum to vertebra L1; 31 segments; conduction pathway and reflex center
Autonomic NS	Sympathetic (thoracolumbar): heart rate and breathing up, blood to skeletal muscle, digestion down, more sweat. Parasympathetic (craniosacral, vagus nerve): rest and digest

Physiology essentials

- **Three functions:** sensory input from receptors → integration in the brain → motor output to effectors (muscles and glands).
- **Resting potential:** about −70 mV inside a typical neuron, set mainly by K^+ leaking out, with the gradients kept up by the Na^+/K^+ pump (ATP).
- **Action potential:** at threshold (about −55 mV) voltage-gated Na^+ channels open and Na^+ rushes in (depolarization); K^+ then flows out (repolarization).
- **Autonomic transmitters:** preganglionic fibers of both divisions release ACh; most sympathetic postganglionic fibers release norepinephrine; parasympathetic postganglionic fibers release ACh.
- **Repair:** peripheral fibers have a neurilemma that helps them regenerate; CNS fibers lack it and do not regenerate.

Special sense	Receptors and pathway
Taste (gustation)	Taste buds detect 5 basic tastes: sweet, salty, sour, bitter and savory (umami)
Smell (olfaction)	Olfactory neurons high in the nasal cavity send axons through the cribriform plate to the olfactory bulb; the only sense that reaches the cortex without a thalamic relay
Hearing (audition)	Eardrum and ossicles pass sound to the cochlea; hair cells of the organ of Corti respond, high pitches near the base, low near the apex; range about 20 to 20,000 Hz
Balance (equilibrium)	Utricle and saccule (otoliths): head position and linear acceleration; semicircular canals: rotation; vestibulocochlear nerve
Vision	Retina: rods for dim light, 3 kinds of cones for color; the fovea gives the sharpest vision; the optic disc is the blind spot

CLINICAL NOTE Stroke

Blood flow to part of the brain is lost and brain cells start dying within minutes. About 80% of strokes are ischemic (a clot); the rest are hemorrhagic (a vessel bursts). Warning signs come on suddenly: numbness or weakness of the face, arm or leg, especially on one side; confusion or trouble speaking; vision loss; trouble walking or balancing; severe headache with no known cause. Call 911 right away: quick treatment can save a life and improve recovery. Source: MedlinePlus, *Stroke* (consulted Oct. 29, 2026).

TRAP
The thalamus relays every sense to the cortex **except smell**.

TRAP
Dendrites carry impulses **toward** the cell body; the axon carries them **away**.

TRAP
7 cervical vertebrae but 8 pairs of cervical nerves: 31 spinal nerve pairs in all (8 C, 12 T, 5 L, 5 S, 1 Co).

Endocrine System

Gland	Hormones and actions
Hypothalamus	Releasing and inhibiting hormones reach the anterior pituitary through the hypophyseal portal veins; its neurons make ADH and oxytocin
Anterior pituitary	GH (growth, protein synthesis), TSH (thyroid), ACTH (adrenal cortex, cortisol), FSH (egg development, sperm production), LH (mid-cycle surge triggers ovulation; makes the testes produce testosterone; at puberty, signals the ovaries to make estrogen), prolactin (milk production)
Posterior pituitary	Stores and releases ADH (water reabsorption by kidney tubules) and oxytocin (uterine contraction, milk ejection); controlled by nerve signals
Pineal gland	Melatonin: daily body rhythms, reproductive development
Thyroid	T_4 (about 95% of active thyroid hormone) and T_3, both need iodine; calcitonin from parafollicular cells lowers blood Ca^{2+}
Parathyroids	4 small glands on the back of the thyroid; PTH raises blood Ca^{2+} and is its main regulator
Adrenal cortex	Steroid hormones: aldosterone (keeps Na^+ and water), cortisol (raises blood glucose), small amounts of sex hormones
Adrenal medulla	Epinephrine and norepinephrine, released on sympathetic stimulation during stress
Pancreatic islets	Alpha cells release glucagon when glucose is low; beta cells release insulin when glucose is high

Physiology essentials

- **Ductless glands:** hormones go straight into the blood and reach the whole body, but act only on cells that carry a matching receptor.
- **Two chemical classes:** protein-based hormones (most of them) bind surface receptors and act fast; steroids (adrenal cortex, sex hormones) bind receptors inside the cell and act more slowly, through protein synthesis.
- **Control:** negative feedback for most hormones; hormonal stimuli (a tropic hormone from one gland makes another gland secrete); neural stimuli (e.g., sympathetic nerves and the adrenal medulla).
- **Feedback axis:** TRH (hypothalamus) → TSH (anterior pituitary) → thyroid hormones; high thyroid hormone levels lower TRH and TSH.
- **Iodine:** without enough iodine the thyroid cannot make T_4 and T_3 and may enlarge (goiter).

CLINICAL NOTE Low blood sugar (hypoglycemia)
Blood glucose below 70 mg/dL (3.9 mmol/L). In diabetes it can follow too much insulin in the blood, or exercise in people taking insulin or some other diabetes medicines. Signs: shaking, sweating, fast heartbeat, hunger, irritability, headache, blurred vision, unclear thinking; severe episodes cause seizures or loss of consciousness, and some people feel no warning at all. Treatment uses fast sugar (a sugary drink or food, or glucose tablets) per protocol; an episode that does not improve, or a person who becomes less alert, needs

emergency care. Source: MedlinePlus, *Low blood sugar* (consulted Oct. 29, 2026).

⚠ TRAP
ADH and oxytocin are **made in the hypothalamus**; the posterior pituitary only stores and releases them.

⚠ TRAP
Ca^{2+}: PTH **raises** it, calcitonin **lowers** it. Glucose: glucagon (alpha cells) and cortisol **raise** it, insulin (beta cells) **lowers** it.

⚠ TRAP
Steroid hormones act **inside** the cell (slower); protein hormones act on surface receptors (faster).

Cardiovascular System

Key structure	Function
Right atrium	Receives deoxygenated blood from the systemic veins
Right ventricle	Pumps blood through the pulmonary semilunar valve into the pulmonary trunk and on to the lungs
Left atrium	Receives oxygenated blood from the pulmonary veins
Left ventricle	Pumps blood through the aortic semilunar valve into the aorta and the systemic circuit
AV valves	Between atrium and ventricle: tricuspid on the right, bicuspid (mitral) on the left
Coronary arteries	Right and left, branches of the ascending aorta; supply the myocardium, which needs a constant flow of oxygen
Conduction system	SA node (pacemaker; intrinsic rate 70–80/min per SEER, without nerve input; normal adult resting heart rate 60–100/min) → AV node → AV bundle → bundle branches → conduction myofibers (Purkinje fibers)
Arteries and arterioles	Carry blood away from the heart; arterioles regulate flow into the capillaries
Capillaries	Exchange between blood and tissue cells by diffusion, filtration and osmosis
Veins	Carry blood toward the heart; medium and large veins have valves, most important in the arms and legs
Blood	About 5 L in an adult, 8% of body weight: formed elements suspended in plasma
Erythrocytes	Red cells: carry oxygen; normally last about 120 days

Physiology essentials

- **Route:** body → right atrium → right ventricle → lungs → left atrium → left ventricle → aorta → body. The pulmonary circuit picks up oxygen; the systemic circuit delivers it.
- **Cardiac cycle:** systole = contraction, diastole = relaxation; one cycle lasts about 0.8 s at a normal heart rate. Heart sounds come from valves closing; abnormal sounds are murmurs.
- **Output:** a normal adult heart pumps about 5 L/min at rest.
- **Blood pressure:** systolic (ventricular contraction) over diastolic (relaxation). It rises with cardiac output, blood volume, peripheral resistance and viscosity.
- **Pulse pressure** = systolic − diastolic. Exercise values: 120/80 mm Hg gives 40 mm Hg.

- **Venous return** relies on skeletal muscle action, breathing movements and venous smooth muscle, not directly on ventricular contraction.

CLINICAL NOTE Heart failure
When the heart pumps poorly, fluid builds up: in the lungs (crackles, shortness of breath on exertion or lying down) and in the liver, abdomen, feet and ankles (swelling). Failure can affect the left side, the right side or both. Source: MedlinePlus, *Heart failure* (consulted Oct. 29, 2026).

TRAP
Arteries leave the heart, veins return to it, whatever their oxygen: **pulmonary arteries** carry low-oxygen blood and **pulmonary veins** oxygenated blood.

TRAP
Tricuspid on the right, **bicuspid (mitral)** on the left. The semilunar valves sit at the exits (pulmonary, aortic), not between chambers.

TRAP
The SA node sets the rhythm on its own; outside regulating factors only speed it up or slow it down.

Lymphatic and Immune Systems

Key structure	Function
Lymph capillaries	Blind-ended sacs in the tissue spaces: interstitial fluid enters but cannot flow back out
Lymphatic vessels	Thin walls and valves, like veins; they carry fluid only away from the tissues
Right lymphatic duct	Drains lymph from the upper right quadrant of the body
Thoracic duct	Drains lymph from the rest of the body
Lymph nodes	Filter lymph; their lymphocytes destroy invading organisms
Spleen	Upper left abdomen: filters blood (white pulp: lymphocytes; red pulp: blood and macrophages), removes old red cells with the liver, stores blood
Thymus	Behind the sternum: maturation of T lymphocytes; secretes thymosin; large in children, small in older adults
Bone marrow	Forms the blood cells, lymphocytes included
B lymphocytes	Become plasma cells that produce antibodies
T lymphocytes	Attack antigens directly and release cytokines that control the immune response

Physiology essentials

- **Three jobs of the lymphatic system:** return excess interstitial fluid to the blood; absorb fats and fat-soluble vitamins; defend against microorganisms.
- **Innate immunity:** barriers that act first: skin, mucus, cough reflex, enzymes in tears and skin oils, stomach acid. Inflammation walls off the invader and draws phagocytes.
- **Acquired immunity:** builds up after exposure to a specific **antigen** (usually a protein on a cell, virus or microbe). Vaccines produce it without the infection.
- **Antibodies** bind one specific antigen and make it easier for immune cells to destroy.
- **Passive immunity:** ready-made antibodies, e.g., from mother to fetus across the placenta. Immediate but short: an infant's maternal antibodies disappear between 6 and 12 months.

CLINICAL NOTE Life without a spleen
After the spleen is removed, infections become more likely. Vaccination is reviewed before the operation, with shots against pneumococcus, meningococcus, *Haemophilus influenzae*, influenza and COVID-19, and discussed again with the care team afterward. Source: MedlinePlus, *Spleen removal* (consulted Oct. 29, 2026).

TRAP
T cells mature in the **thymus**. Antibodies come from **plasma cells** (activated B cells), not from T cells.

TRAP
Passive immunity is **fast but temporary**; vaccines give lasting **acquired** immunity.

TRAP
Lymph flows **one way**, from the tissues toward the blood. The right lymphatic duct drains only the upper right quadrant; the thoracic duct drains the rest.

Respiratory System

Key structure	Function
Nose and sinuses	Nose hairs trap large inhaled particles; the paranasal sinuses produce mucus
Pharynx	Serves both systems: air from the nose, air, food and water from the mouth; nasopharynx, oropharynx, laryngopharynx
Larynx	Voice box; the glottis is the opening between the vocal cords; 9 cartilages, the thyroid cartilage is the Adam's apple; vocal cords vibrate to make sound; the epiglottis keeps food out
Trachea	Hyaline cartilage keeps it from collapsing; divides into the right and left primary bronchi at vertebra T5
Bronchial tree	Branches into smaller and smaller tubes; cartilage dwindles and is absent in the smallest bronchioles
Alveoli	Simple squamous epithelium: rapid diffusion of gases; site of gas exchange
Lungs	Right lung 3 lobes, left lung 2 lobes
Pleura	Visceral pleura on the lung, parietal pleura lining the chest; the pleural cavity between them holds a thin film of serous fluid
Diaphragm	Contracts on inspiration and enlarges the thoracic cavity

Physiology essentials

- **Pressure rule:** air flows from higher to lower pressure.
- **Inspiration (active):** the diaphragm contracts, chest volume rises, pressure in the alveoli falls below the atmosphere, air flows in.
- **Expiration at rest:** the diaphragm relaxes and elastic tissue recoils, chest volume falls, air flows out.
- **Rate:** an average adult takes 12–15 breaths/min at rest.
- **Respiration** is the whole chain of O_2 and CO_2 exchange: ventilation, exchange in the alveoli, transport by the blood, exchange with the tissue cells.

CLINICAL NOTE Collapsed lung (pneumothorax)
Air escapes from the lung into the space between the lung and the chest wall. The trapped air presses on the lung, which can no longer expand fully. Typical signs: sharp chest or shoulder pain made worse by a deep breath or a cough, and shortness of breath. Source: MedlinePlus, *Collapsed lung (pneumothorax)* (consulted Oct. 29, 2026).

TRAP
Larynx (voice box, air only, guarded by the epiglottis) vs **pharynx** (throat, shared by air and food).

TRAP
Right lung **3** lobes, left lung **2**. The trachea splits at vertebra T5.

TRAP
Quiet expiration is passive. Only inspiration needs muscle contraction at rest.

Digestive System

Key structure	Function
Mouth	Chewing (mechanical digestion); saliva has an enzyme that starts starch digestion; 20 primary teeth, 32 permanent teeth
Esophagus	Moves food to the stomach by peristalsis
Stomach	Acid and enzymes break down proteins; parietal cells secrete HCl and intrinsic factor; endocrine cells secrete gastrin; mixing movements make chyme
Pyloric sphincter	Relaxes to let chyme pass into the small intestine
Small intestine	Duodenum, jejunum, ileum: absorbs most nutrients; plicae circulares, villi and microvilli enlarge the surface; secretes CCK and secretin
Liver	Makes bile; also synthesizes plasma proteins, stores nutrients, detoxifies, and handles carbohydrate, lipid and protein metabolism
Gallbladder	Stores bile between meals
Pancreas	Exocrine acini: amylase, trypsin, peptidase, lipase; endocrine islets: insulin and glucagon
Large intestine	Colon, rectum, anal canal: no digestive enzymes; absorbs water and electrolytes; its bacteria make vitamin K

Physiology essentials

- **Sequence:** ingestion → mechanical digestion (chewing, mixing) → chemical digestion (enzymes speed up hydrolysis) → absorption → elimination (defecation).
- **Peristalsis** moves food along the whole GI tract.
- **Carbohydrates:** salivary enzyme, then pancreatic amylase and intestinal enzymes. **Proteins:** stomach acid and enzymes, then pancreatic trypsin and peptidases. **Fats:** bile salts emulsify, pancreatic lipase digests.
- **Hormones:** gastrin regulates gastric activity; secretin and CCK, made in the small intestine, control pancreatic secretion.
- **Absorption:** the small intestine absorbs most nutrients; the large intestine takes back more water and electrolytes.

CLINICAL NOTE Vitamin B_{12} and intrinsic factor

Stomach acid and gastric protease free B_{12} from food; it binds intrinsic factor from the parietal cells and is absorbed in the distal ileum. Pernicious anemia (autoimmune attack on parietal cells) and surgery that removes part of the stomach can cause deficiency. Source: NIH Office of Dietary Supplements, *Vitamin B12, Fact Sheet for Health Professionals* (consulted Oct. 29, 2026).

TRAP

The **liver makes** bile; the **gallbladder only stores** it. Bile emulsifies fat; the enzyme that digests fat is **lipase**.

TRAP

Most absorption happens in the **small intestine**, not the stomach. The large intestine secretes **no digestive enzymes**.

Urinary System and Fluid Balance

Key structure	Function
Kidneys	Retroperitoneal; outer cortex, inner medulla; the renal pelvis collects urine as it forms
Nephron	Functional unit, over 1 million per kidney: renal corpuscle (glomerulus in the glomerular capsule) + renal tubule (with the nephron loop)
Glomerulus	Its thin capillary walls let small molecules, wastes and fluid (mostly water) pass into the tubule
Renal tubule	Returns most of the filtered water and solutes to the blood; removes excess acid
Juxtaglomerular apparatus	Monitors blood pressure and secretes renin
Collecting duct	Carries urine from the nephrons to the calyces and renal pelvis
Ureters	Small tubes, about 25 cm long; peristalsis moves urine from the renal pelvis to the bladder
Urethra	Carries urine out of the body; in males, also the passage for sperm

Physiology essentials

- **Main job:** keep the volume and composition of body fluids within normal limits: water excretion, electrolytes, blood pH. The kidneys also secrete erythropoietin (red cell production) and renin (blood pressure).
- **Urine formation:** the glomeruli filter about 150 quarts/day of blood; the tubules return most of it, so only 1–2 quarts/day becomes urine.
- **ADH** (posterior pituitary): kidney tubules reabsorb **water**.
- **Aldosterone** (adrenal cortex): kidneys reabsorb **sodium and water** and release **potassium**; blood pressure rises.
- **Reference ranges (MedlinePlus):** Na^+ 135–145 mEq/L, K^+ 3.7–5.2 mEq/L. Ranges vary slightly among labs: use your program's.

CLINICAL NOTE High potassium
Kidneys that do not work well may fail to remove enough potassium. High potassium decreases heart muscle activity: palpitations, a slow, weak or irregular pulse, and in severe cases cardiac arrest. Source: MedlinePlus, *High potassium level* and *Potassium blood test* (consulted Oct. 29, 2026).

TRAP
ADH saves **water only**. Aldosterone saves **sodium** (water follows) and gets rid of **potassium**.

TRAP
Renin comes from the kidney (juxtaglomerular apparatus); **aldosterone** from the adrenal cortex.

Reproductive System

Key structure	Function
Testes	Seminiferous tubules make sperm (spermatogenesis); interstitial (Leydig) cells make male sex hormones; Sertoli cells support and nourish
Scrotum	Holds the testes about 3 °C below body temperature
Epididymis	Sperm mature and become fertile here; mature sperm are stored in its tail
Ductus deferens	Stores sperm near the epididymis; peristalsis moves them toward the ejaculatory duct
Ejaculatory duct	Receives seminal vesicle secretions, passes through the prostate and empties into the urethra
Ovaries	Oogenesis; granulosa cells of the follicle secrete estrogen; the corpus luteum secretes progesterone and some estrogen
Uterine (fallopian) tubes	Fimbriae sweep the oocyte in; fertilization usually happens here (oocyte fertile 24–48 h)
Uterus	Lined with the endometrium; site of menstruation and of pregnancy
Vagina	Fibromuscular tube from the cervix of the uterus to the outside

Physiology essentials

- **Axis:** hypothalamus (GnRH) → anterior pituitary (FSH, LH) → testes or ovaries.
- **Meiosis:** one primary spermatocyte gives 4 spermatids; one primary oocyte gives **one** fertilizable cell, and the polar bodies degenerate.
- **Ovarian cycle:** follicular phase → ovulation (a mid-cycle rise in LH releases the oocyte) → luteal phase. Without fertilization the corpus luteum works about 10 days, then regresses.
- **Uterine cycle:** menstrual phase → proliferative phase (endometrium repaired) → secretory phase (glands and blood vessels grow).
- **Menopause:** the cycles stop; changing hormone levels cause its symptoms.

CLINICAL NOTE Ectopic pregnancy

The most common site is the fallopian tube. Risks include tubal birth defects, a previous ectopic pregnancy, scarring from past infections or surgery, and sexually transmitted infections. Rupture causes bleeding and shock: an emergency. Source: MedlinePlus, *Ectopic pregnancy* (consulted Oct. 29, 2026).

TRAP

Sperm are **made** in the seminiferous tubules but **mature** in the epididymis.

TRAP

Ovulation follows the mid-cycle rise in **LH**, not FSH. The corpus luteum secretes **progesterone** (and some estrogen).